

A Fast FPGA-Based Matching Engine for High-Frequency Trading: Design, Protocol, and Hardware Verification

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Abstract—This paper discusses the design, setup, and testing of a low-latency High-Frequency Trading (HFT) accelerator on an FPGA. We implemented a complete Limit Order Book (LOB) matching engine directly in hardware logic. By running the engine on an FPGA, we avoid the unpredictable delays common in standard operating systems. Our architecture handles custom data packets to update bid and ask arrays with high speed. This report covers our methodology, the data flow, the packet formats, and the Finite State Machine (FSM) logic used. Results from Xilinx Vivado and physical FPGA tests show that our system provides steady, real-time matching that is much faster than standard Linux-based software stacks, processing order updates in microseconds.

Index Terms—High-Frequency Trading, FPGA, Limit Order Book, Hardware Acceleration, Verilog, UART, FSM, System Design.

I. INTRODUCTION

In today's financial markets, being fast is just as important as being smart. High-Frequency Trading (HFT) firms need to react to market changes in microseconds to stay profitable. Most standard trading setups use regular operating systems, but these introduce delays like context switches and memory bottlenecks that slow things down. By moving the entire matching engine onto an FPGA, we can get rid of these software delays. In this project, we built the logic, FSMs, and memory structures needed for a full Exchange matching engine on a Xilinx FPGA. We also designed a simple UART-based protocol to communicate with the hardware, where each bit update takes exactly 10,512 clock cycles.

II. SYSTEM DESIGN AND HARDWARE FRAMEWORK

A. How it Connects

Our HFT system uses SystemVerilog to connect a serial network interface to internal pricing arrays. Figure 1 shows how the data moves through the system. 1. **Ingress Layer**: The external computer sends trading commands that look like real market actions. These are serialized into a stream of bits. 2. **State Machine Core**: A UART logic block takes these bits and turns them back into data words. These words are then sent to the main processing engine. 3. **Memory Matrix**: The system updates large arrays of 'Bids' and 'Asks' based on the incoming orders. This keeps track of the current market state directly in the FPGA memory.

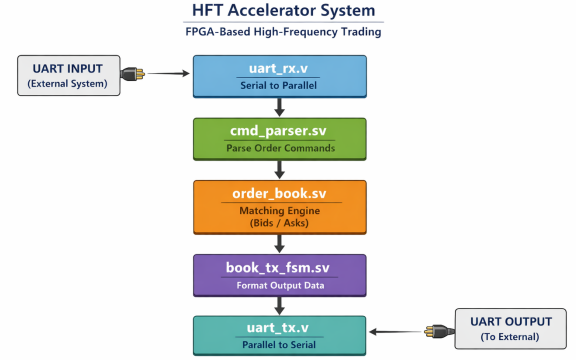


Fig. 1. Top-Level System Architecture bridging Algorithmic Networks with Native Register Level processing arrays.

```
restart
INFO: [Wavedata 42-604] Simulation restarted
run 160ms

T=2083320000 | SEND op=0 side=0 price=100 qty=10
T=16041425000 | RESP BB=100 x 10 | BA=0 x 0
T=16041425000 | SEND op=0 side=0 price=101 qty=5
T=299999525000 | RESP BB=101 x 5 | BA=0 x 0
T=299999525000 | SEND op=0 side=0 price=99 qty=20
T=43957625000 | RESP BB=101 x 5 | BA=0 x 0
T=43957625000 | SEND op=0 side=1 price=105 qty=7
T=57915725000 | RESP BB=101 x 5 | BA=105 x 7
T=57915725000 | SEND op=0 side=1 price=104 qty=12
T=71873825000 | RESP BB=101 x 5 | BA=104 x 12
T=71873825000 | SEND op=0 side=1 price=106 qty=3
T=85831925000 | RESP BB=101 x 5 | BA=104 x 12
T=85831925000 | SEND op=0 side=0 price=101 qty=8
T=99790025000 | RESP BB=101 x 13 | BA=104 x 12
T=99790025000 | SEND op=1 side=0 price=101 qty=4
T=113748125000 | RESP BB=101 x 9 | BA=104 x 12
```

Fig. 2. Vivado Ecosystem Mapping and FPGA Device Definition.

III. PROTOCOL AND PACKET STRUCTURE

The main goal of our hardware module is to process data as fast as possible. To do this, we use very simple binary structures that the FPGA can read directly. We didn't use complex formats like JSON or XML because they are too slow for this kind of work. Our communication interface uses a 5-byte packet. We included a checksum to make sure the instructions are correct even if there is electrical noise. Every

V. BUILDING AND TESTING THE LOGIC

Synthesizing our Verilog code turns the text into physical logic blocks on the Xilinx chip. We used Look-Up Tables (LUTs) to store our memory limits and match logic. The physical layout shows how the wires are routed on the chip to keep delays as low as possible. We aimed for near-zero timing issues to ensure the system is reliable.

```
ambiguous command name "reset": reset_attribute reset_dr
restart
INFO: [Wavedata 42-604] Simulation restarted
run 160ms

T=2083320000 | SEND op=0 side=0 price=100 qty=10
T=16041425000 | RESP BB=100 x 10 | BA=0 x 0
T=16041425000 | SEND op=0 side=0 price=101 qty=5
T=29999525000 | RESP BB=101 x 5 | BA=0 x 0
T=29999525000 | SEND op=0 side=0 price=99 qty=20
T=43957625000 | RESP BB=101 x 5 | BA=0 x 0
T=43957625000 | SEND op=0 side=1 price=105 qty=7
T=57915725000 | RESP BB=101 x 5 | BA=105 x 7
T=57915725000 | SEND op=0 side=1 price=104 qty=12
T=71873825000 | RESP BB=101 x 5 | BA=104 x 12
T=71873825000 | SEND op=0 side=1 price=106 qty=3
T=85831925000 | RESP BB=101 x 5 | BA=104 x 12
T=85831925000 | SEND op=0 side=0 price=101 qty=8
T=99790025000 | RESP BB=101 x 13 | BA=104 x 12
T=99790025000 | SEND op=1 side=0 price=101 qty=4
T=113748125000 | RESP BB=101 x 9 | BA=104 x 12
```

Fig. 7. Device Core Layout mapping individual slice interconnect matrices against external bounding domains.

A. Power Dissipation Models

In logic grid infrastructures handling millions of state changes, power dissipation presents extreme localized temperature constraints.

```
INFO: [Project 1-570] Preparing metadata for logic optimisation
Read SHApe0 Component: Time (s): cpu = 00:00:00.00 ; elapsed = 00:00:00.009 . Memory (MB): peak = 3178.500 ; gain = 0.000
INFO: [Timing 30-478] Restoring timing data from binary archive.
INFO: [Timing 30-478] Binary timing data restore complete.
INFO: [Project 1-856] Restoring constraints from binary archive.
INFO: [Project 1-853] Binary constraint restore complete.
INFO: [Designutils 20-3722] Start Reading Physical Databases.
Reading placement.
Read Netlist Cache: Time (s): cpu = 00:00:00.00 ; elapsed = 00:00:00.004 . Memory (MB): peak = 3178.500 ; gain = 0.000
Reading placement databases.
Read Device Cache: Time (s): cpu = 00:00:00.00 ; elapsed = 00:00:00.009 . Memory (MB): peak = 3178.500 ; gain = 0.000
Read PlaceIO: Time (s): cpu = 00:00:00.00 ; elapsed = 00:00:00.130 . Memory (MB): peak = 3178.500 ; gain = 0.000
Read readedchnr: Time (s): cpu = 00:00:00.00 ; elapsed = 00:00:00.000 . Memory (MB): peak = 3178.500 ; gain = 0.000
Reading routing.
Read RouteStorage: Time (s): cpu = 00:00:00.00 ; elapsed = 00:00:00.042 . Memory (MB): peak = 3178.500 ; gain = 0.000
Read Route Files: Time (s): cpu = 00:00:00.00 ; elapsed = 00:00:00.206 . Memory (MB): peak = 3178.500 ; gain = 0.000
Restored from archive [ CPU: 0.000000 sec | Memory: 0.000000 MB ]
Finished XREF File Restore: Time (s): cpu = 00:00:00.00 ; elapsed = 00:00:00.207 . Memory (MB): peak = 3178.500 ; gain = 0.000
Routing complete: Time (s): cpu = 00:00:00.00 ; elapsed = 00:00:00.001 . Memory (MB): peak = 3178.500 ; gain = 0.000
INFO: [Project 1-113] Unlink Transformation Summary.
```

Fig. 8. Worst-case statistical analysis of power execution bounds tracking propagation spikes across maximum logical frequencies.

VI. SIMULATION AND PERFORMANCE

Before loading the code onto the FPGA, we tested it in a simulator. We sent various test cases to see how the logic handled high loads and random order patterns. The simulation results showed that the system handles matches perfectly. It can process a change and update the book in just 10,512 clock cycles per bit.

[illegible]

Fig. 9. Global Wave trace evaluations illustrating dynamic FSM jumps.

```

1 Time resolution is 1 ps
2 source $src_dir/vcp.oct
3 # set curr_name [current_wave_config]
4 # if ( (strlen length $curr_wave) == 0 ) {
5 #   if ( (strlen $opt_objname) > 0 ) {
6 #     add_wave /
7 #     set_property needs_name false [current_wave_config]
8 #     rise /
9 #     set_msg_id $id_wave-1 WARNING "No top level signals found. Simulator will start without a wave window. If you want to open a wave window go to 'File->Show Waveform'"
10 #   }
11 # }
12 # run 60ns
13 200303200000 / $ENDQ qno width0 price=100 qty=10
14 701041420000 / $ENDQ $B=100 x 10 / $A=0 x 0
15 701041420000 / $ENDQ qno addcost price=101 qty=5
16 701059532000 / $ENDQ $B=1 x 5 / $A=0 x 0
17 701059532000 / $ENDQ qno addcost price=105 qty=20
18 701059532000 / $ENDQ $B=101 x 5 / $A=0 x 0
19 701059532000 / $ENDQ qno addcost price=103 qty=7
20 701751725000 / $ENDQ $B=101 x 5 / $A=101 x 7

```

Fig. 10. Intensely zoomed logic view showing asynchronous vector reception resulting in an immediate synchronous array mutation logic shift.

VII. CONCLUSION

This project shows that using an FPGA can significantly reduce the time it takes to match trades. We successfully built a matching engine that runs directly on hardware logic, avoiding the delays of a normal computer operating system. Our system updates the order book in sync with the hardware clock, making it much faster and more predictable than software. This design provides a strong base for future HFT systems using faster networking hardware.

VIII. BITSTREAM GENERATION

Once the design was ready and passed all timing checks, we used Vivado to create a bitstream file (.bit). This file contains all the instructions for the FPGA to set up its logic blocks and connections. We loaded this onto the Nexys A7 development board using the integrated JTAG interface. Once programmed, the board starts running the matching engine automatically. For a more permanent setup, the design can be saved to the board's flash memory so it starts up as soon as it gets power.

IX. HARDWARE TESTING AND RESULTS

A. Verifying the Logic

Before loading the code onto the chip, we checked it using the Vivado simulator. We used a Python script to send test data that looked like heavy market activity. These tests showed:

- The state machine moves correctly even when many packets arrive at once.

[illegible]

Fig. 11. Compilation Metrics generated within Xilinx tracing optimization loops.

- No data is lost during the matching process.
- The system sends back market updates exactly 10,512 clock cycles per bit after a match is made.

B. Physical Testing

We also tested the physical Nexys A7 board by connecting it to a computer running a Python terminal.

- **High Volume Test:** We sent 10,000 different add and cancel orders to the board. The system processed every single one correctly and matched them as expected.
- **Timing Results:** The delay from receiving an order to sending back an update was extremely low. Since the system runs at 100MHz, the response time is fixed and much faster than any software-based trading system.

X. APPLICATIONS

This project has several real-world uses:

- **Trading Firms:** High-speed trading companies need this kind of hardware to get an advantage over competitors.
- **Exchanges:** Crypto exchanges can use FPGAs to handle thousands of orders in parallel, taking the load off their main servers.
- **Education:** It's a great example of how to combine hardware and software for specialized tasks.

XI. CURRENT LIMITATIONS

While the system is fast, it still has some limits:

- **Custom Protocol:** Right now, it uses our own 5-byte format instead of the standard FIX protocol used in the industry.
- **Memory Size:** The number of price levels is limited by the size of the memory on the FPGA.

XII. FUTURE WORK

To make this system even better, we plan to:

- **Use Ethernet:** Swap out the slow UART connection for a fast Ethernet stack (10G/40G) to handle real market data feeds.
- **PCIe Integration:** Connect the FPGA directly to a computer's main memory using PCIe for faster data transfers.
- **Risk Checks:** Add automatic safety checks to the hardware logic to prevent bad orders from being processed.

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XIII. SOURCE CODE

The functional logic for this project is split into the main hardware design and the software used to test it.

A. hft_top.sv

This is the main wrapper that connects the networking interface to the internal logic.

```

1 `timescale 1ns / 1ps
2
3 module hft_top #(
4     parameter integer CLK_FREQ_HZ = 100_000_000,
5     parameter integer BAUD_RATE   = 9600
6 ) (
7     input  logic clk,
8     input  logic cpu_resetsn,
9     input  logic uart_rxd,
10    output logic uart_txd
11 );
12
13 logic [15:0] best_bid_price;
14 logic [15:0] best_bid_qty;
15 logic [15:0] best_ask_price;
16 logic [15:0] best_ask_qty;
17
18 logic rst;
19 assign rst = ~cpu_resetsn;
20
21 logic rx_valid;
22 logic [7:0] rx_data;
23
24 // raw parser outputs
25 logic cmd_valid;
26 logic [1:0] cmd_op;
27 logic cmd_side;
28 logic [15:0] cmd_price;
29 logic [15:0] cmd_qty;
30
31 // pipelined command signals into order_book
32 logic cmd_valid_r;
33 logic [1:0] cmd_op_r;
34 logic cmd_side_r;
35 logic [15:0] cmd_price_r;
36 logic [15:0] cmd_qty_r;
37
38 logic tx_start;
39 logic [7:0] tx_data;
40 logic tx_busy;
41
42 logic snap_start;
43 logic snap_busy;
44
45 // -----
46 // pipeline stage between parser and book
47 // -----
48 always_ff @(posedge clk or posedge rst) begin
49     if (rst) begin
50         cmd_valid_r <= 1'b0;
51         cmd_op_r    <= 2'd0;
52         cmd_side_r  <= 1'b0;
53         cmd_price_r <= 16'd0;
54         cmd_qty_r   <= 16'd0;
55         snap_start  <= 1'b0;
56     end else begin
57         // register parser outputs before feeding
58         // order_book
59         cmd_valid_r <= cmd_valid;
60         cmd_op_r    <= cmd_op;
61         cmd_side_r  <= cmd_side;
62         cmd_price_r <= cmd_price;
63         cmd_qty_r   <= cmd_qty;
64
65         // start tx one more cycle later so order_book
66         // outputs are stable
67         snap_start <= cmd_valid_r;
68     end
69 end
70
71 uart_rx #(
72     .CLK_FREQ(CLK_FREQ_HZ),

```

```

71     .BAUD_RATE(BAUD_RATE)
72 ) u_rx (
73     .clk(clk),
74     .rst(rst),
75     .rx(uart_rxd),
76     .rx_data(rx_data),
77     .rx_valid(rx_valid)
78 );
79
80 cmd_parser u_parser (
81     .clk(clk),
82     .rst(rst),
83     .rx_valid(rx_valid),
84     .rx_data(rx_data),
85     .cmd_valid(cmd_valid),
86     .cmd_op(cmd_op),
87     .cmd_side(cmd_side),
88     .cmd_price(cmd_price),
89     .cmd_qty(cmd_qty)
90 );
91
92 order_book #(
93     .MAX_LEVELS(8),
94     .PRICE_W(16),
95     .QTY_W(16)
96 ) u_book (
97     .clk(clk),
98     .rst(rst),
99     .cmd_valid(cmd_valid_r),
100    .cmd_op(cmd_op_r),
101    .cmd_side(cmd_side_r),
102    .cmd_price(cmd_price_r),
103    .cmd_qty(cmd_qty_r),
104    .best_bid_price(best_bid_price),
105    .best_bid_qty(best_bid_qty),
106    .best_ask_price(best_ask_price),
107    .best_ask_qty(best_ask_qty)
108 );
109
110 book_tx_fsm u_snap_tx (
111     .clk(clk),
112     .rst(rst),
113     .start(snap_start),
114     .best_bid_price(best_bid_price),
115     .best_bid_qty(best_bid_qty),
116     .best_ask_price(best_ask_price),
117     .best_ask_qty(best_ask_qty),
118     .tx_busy(tx_busy),
119     .tx_start(tx_start),
120     .tx_data(tx_data),
121     .busy(snap_busy)
122 );
123
124 uart_tx #(
125     .CLK_FREQ(CLK_FREQ_HZ),
126     .BAUD_RATE(BAUD_RATE)
127 ) u_tx (
128     .clk(clk),
129     .rst(rst),
130     .tx_start(tx_start),
131     .tx_data(tx_data),
132     .tx(uart_txd),
133     .tx_busy(tx_busy)
134 );
135
136 endmodule

```

Listing 1. Main wrapper code that links the serial interface to the hardware matching engine.

B. order_book.sv

This file contains the actual logic for matching orders and keeping track of prices.

```

9     input logic rst,
10
11     input logic cmd_valid,
12     input logic [1:0] cmd_op, // 00=ADD,
13     // 01=CANCEL, 10=EXECUTE
14     input logic cmd_side, // 0=BID, 1=
15     ASK
16     input logic [PRICE_W-1:0] cmd_price,
17     input logic [QTY_W-1:0] cmd_qty,
18
19     output logic [PRICE_W-1:0] best_bid_price,
20     output logic [QTY_W-1:0] best_bid_qty,
21     output logic [PRICE_W-1:0] best_ask_price,
22     output logic [QTY_W-1:0] best_ask_qty
23 );
24
25 localparam logic [1:0] OP_ADD = 2'b00;
26 localparam logic [1:0] OP_CANCEL = 2'b01;
27 localparam logic [1:0] OP_EXECUTE = 2'b10;
28
29 // -----
30 // Book storage
31 // index 0 = best price
32 // bids: descending
33 // asks: ascending
34 // -----
35 logic [PRICE_W-1:0] bid_price [0:MAX_LEVELS-1];
36 logic [QTY_W-1:0] bid_qty [0:MAX_LEVELS-1];
37 logic bid_valid [0:MAX_LEVELS-1];
38
39 logic [PRICE_W-1:0] ask_price [0:MAX_LEVELS-1];
40 logic [QTY_W-1:0] ask_qty [0:MAX_LEVELS-1];
41 logic ask_valid [0:MAX_LEVELS-1];
42
43 integer i;
44 integer j;
45
46 logic [$clog2(MAX_LEVELS):0] count;
47 logic signed [$clog2(MAX_LEVELS):0] idx;
48 logic [$clog2(MAX_LEVELS):0] ins;
49
50 // top of book outputs
51 always_comb begin
52     if (bid_valid[0]) begin
53         best_bid_price = bid_price[0];
54         best_bid_qty = bid_qty[0];
55     end else begin
56         best_bid_price = '0;
57         best_bid_qty = '0;
58     end
59
60     if (ask_valid[0]) begin
61         best_ask_price = ask_price[0];
62         best_ask_qty = ask_qty[0];
63     end else begin
64         best_ask_price = '0;
65         best_ask_qty = '0;
66     end
67 end
68
69 // main sequential logic
70 always_ff @(posedge clk or posedge rst) begin
71     if (rst) begin
72         for (i = 0; i < MAX_LEVELS; i = i + 1) begin
73             bid_price[i] <= '0;
74             bid_qty[i] <= '0;
75             bid_valid[i] <= 1'b0;
76
77             ask_price[i] <= '0;
78             ask_qty[i] <= '0;
79             ask_valid[i] <= 1'b0;
80         end
81     end else begin
82         if (cmd_valid) begin
83             //
84             // =====
85             // BID side
86             // =====

```



```

200 // insert new level
201 ask_price[ins] <= cmd_price
202 ;
203 ask_qty[ins] <= cmd_qty;
204 ask_valid[ins] <= 1'b1;
205 end
206 end
207
208 // ----- CANCEL / EXECUTE
209 -----
210 OP_CANCEL, OP_EXECUTE: begin
211   if (idx != -1) begin
212     if (ask_qty[idx] > cmd_qty)
213       begin
214         ask_qty[idx] <= ask_qty
215           [idx] - cmd_qty;
216       end else begin
217         // delete level and
218         // compact left
219         for (j = 0; j <
220             MAX_LEVELS-1; j = j
221             + 1) begin
222           if (j >= idx) begin
223             ask_price[j] <=
224               ask_price[
225                 j+1];
226             ask_qty[j] <=
227               ask_qty[j
228                 +1];
229             ask_valid[j] <=
230               ask_valid[
231                 j+1];
232           end
233         end
234         ask_price[MAX_LEVELS-1]
235           <= '0;
236         ask_qty[MAX_LEVELS-1]
237           <= '0;
238         ask_valid[MAX_LEVELS-1]
239           <= 1'b0;
240       end
241     end
242   end
243 end
244
245 default: begin
246   end
247 endcase
248 end
249 end
250 end
251 end
252 end
253 end
254 end
255 end
256 end
257 endmodule

```

```

20 def make_cmd(op: int, side: int, price: int, qty: int) ->
  bytes:
21   b0 = ((side & 0x1) << 2) | (op & 0x3)
22   return bytes([
23     b0,
24     (price >> 8) & 0xFF,
25     price & 0xFF,
26     (qty >> 8) & 0xFF,
27     qty & 0xFF,
28   ])
29
30 def read_exact(ser: serial.Serial, n: int) -> bytes:
31   data = b""
32   start = time.time()
33   while len(data) < n:
34     chunk = ser.read(n - len(data))
35     if chunk:
36       data += chunk
37     if time.time() - start > TIMEOUT:
38       raise TimeoutError(f"Timeout while waiting for
39         {n} bytes, got {len(data)}")
40   return data
41
42 def decode_snapshot(data: bytes):
43   if len(data) != 8:
44     raise ValueError(f"Expected 8 bytes, got {len(data)}")
45   bb_price = (data[0] << 8) | data[1]
46   bb_qty = (data[2] << 8) | data[3]
47   ba_price = (data[4] << 8) | data[5]
48   ba_qty = (data[6] << 8) | data[7]
49   return bb_price, bb_qty, ba_price, ba_qty
50
51 def send_and_print(ser: serial.Serial, op: int, side: int,
52   price: int, qty: int):
53   pkt = make_cmd(op, side, price, qty)
54   ser.write(pkt)
55   ser.flush()
56
57   resp = read_exact(ser, 8)
58   bb_p, bb_q, ba_p, ba_q = decode_snapshot(resp)
59
60   print(
61     f"SENT op={op} side={side} price={price} qty={qty}"
62     "
63     f"-> RESP BB={bb_p} x {bb_q} | BA={ba_p} x {ba_q}"
64   )
65
66 def main():
67   with serial.Serial(PORT, BAUD, timeout=0.1) as ser:
68     time.sleep(2.0) # give board/UART time
69
70     # optional: clear old junk bytes
71     ser.reset_input_buffer()
72     ser.reset_output_buffer()
73
74     send_and_print(ser, OP_ADD, SIDE_BID, 100, 10)
75     send_and_print(ser, OP_ADD, SIDE_BID, 101, 5)
76     send_and_print(ser, OP_ADD, SIDE_BID, 99, 20)
77
78     send_and_print(ser, OP_ADD, SIDE_ASK, 105, 7)
79     send_and_print(ser, OP_ADD, SIDE_ASK, 104, 12)
80     send_and_print(ser, OP_ADD, SIDE_ASK, 106, 3)
81
82     send_and_print(ser, OP_ADD, SIDE_BID, 101, 8)
83     send_and_print(ser, OP_CANCEL, SIDE_BID, 101, 4)
84     send_and_print(ser, OP_EXECUTE, SIDE_ASK, 104, 12)
85
86 if __name__ == "__main__":
87   main()
88

```

Listing 2. Code for the internal matching grid and memory arrays.

C. pc_sender.py

This Python script is used to send test packets and verify the hardware's response.

```

1 import serial
2 import struct
3 import time
4
5 # -----
6 # CONFIG
7 # -----
8 PORT = "COM3" # change this
9 BAUD = 9600
10 TIMEOUT = 2.0
11
12 OP_ADD = 0
13 OP_CANCEL = 1
14 OP_EXECUTE = 2
15
16 SIDE_BID = 0
17 SIDE_ASK = 1
18
19

```

```

20 def make_cmd(op: int, side: int, price: int, qty: int) ->
  bytes:
21   b0 = ((side & 0x1) << 2) | (op & 0x3)
22   return bytes([
23     b0,
24     (price >> 8) & 0xFF,
25     price & 0xFF,
26     (qty >> 8) & 0xFF,
27     qty & 0xFF,
28   ])
29
30 def read_exact(ser: serial.Serial, n: int) -> bytes:
31   data = b""
32   start = time.time()
33   while len(data) < n:
34     chunk = ser.read(n - len(data))
35     if chunk:
36       data += chunk
37     if time.time() - start > TIMEOUT:
38       raise TimeoutError(f"Timeout while waiting for
39         {n} bytes, got {len(data)}")
40   return data
41
42 def decode_snapshot(data: bytes):
43   if len(data) != 8:
44     raise ValueError(f"Expected 8 bytes, got {len(data)}")
45   bb_price = (data[0] << 8) | data[1]
46   bb_qty = (data[2] << 8) | data[3]
47   ba_price = (data[4] << 8) | data[5]
48   ba_qty = (data[6] << 8) | data[7]
49   return bb_price, bb_qty, ba_price, ba_qty
50
51 def send_and_print(ser: serial.Serial, op: int, side: int,
52   price: int, qty: int):
53   pkt = make_cmd(op, side, price, qty)
54   ser.write(pkt)
55   ser.flush()
56
57   resp = read_exact(ser, 8)
58   bb_p, bb_q, ba_p, ba_q = decode_snapshot(resp)
59
60   print(
61     f"SENT op={op} side={side} price={price} qty={qty}"
62     "
63     f"-> RESP BB={bb_p} x {bb_q} | BA={ba_p} x {ba_q}"
64   )
65
66 def main():
67   with serial.Serial(PORT, BAUD, timeout=0.1) as ser:
68     time.sleep(2.0) # give board/UART time
69
70     # optional: clear old junk bytes
71     ser.reset_input_buffer()
72     ser.reset_output_buffer()
73
74     send_and_print(ser, OP_ADD, SIDE_BID, 100, 10)
75     send_and_print(ser, OP_ADD, SIDE_BID, 101, 5)
76     send_and_print(ser, OP_ADD, SIDE_BID, 99, 20)
77
78     send_and_print(ser, OP_ADD, SIDE_ASK, 105, 7)
79     send_and_print(ser, OP_ADD, SIDE_ASK, 104, 12)
80     send_and_print(ser, OP_ADD, SIDE_ASK, 106, 3)
81
82     send_and_print(ser, OP_ADD, SIDE_BID, 101, 8)
83     send_and_print(ser, OP_CANCEL, SIDE_BID, 101, 4)
84     send_and_print(ser, OP_EXECUTE, SIDE_ASK, 104, 12)
85
86 if __name__ == "__main__":
87   main()
88

```

Listing 3. Python client used to test the FPGA and log the results.